

6G PG Class-Lecture-14

Transcript

https://otter.ai/u/YlidrcsoJM8J9Edvmxh_FGZp8EQ?view=summary

Overview

Prof. Neelkanth Kundu discussed the private capacity of quantum channels, essential for secure quantum key distribution. He explained that Alice encodes a message into a quantum code word, which is transmitted over multiple channel uses. The quantum channel is represented by an isometric operator U , which transforms Alice's quantum state into outputs accessible to Bob and Eve. The private capacity is defined as the maximum mutual information between Alice and Bob minus the mutual information between Alice and Eve. For the identity channel, the private capacity is one bit per channel use. The session also covered quantum teleportation, detailing how Alice teleports a qubit to Bob using shared entanglement and classical communication.

Outline

Private Capacity of Quantum Channels

- Prof. Neelkanth Kundu introduces the topic of private capacity, which is crucial for secure communication in quantum key distribution protocols.
- The goal is to securely distribute keys known only to Alice and Bob, with no information leaked to the eavesdropper.
- Alice encodes a private message M into a quantum code word, which is transmitted over multiple uses of the quantum channel.
- The quantum channel is represented by an isometric operator U , which transforms Alice's quantum state into outputs accessible to Bob and Eve.

Defining Quantum Channels and Isometric Operators

- Prof. Neelkanth Kundu explains the concept of the main quantum channel, which characterizes the density matrix accessible to Bob.
- The complementary channel characterizes the density matrix accessible to Eve.
- The combined state of Bob and Eve is denoted by $U \sigma U^\dagger$.
- Partial trace is used to characterize the quantum state accessible to Eve, given by trace over B of $U \sigma U^\dagger$.

Discussion on Quantum States and Dimensions

- Brajesh Kumar and Paramshiv S ask about the dimensions of quantum states and matrices.
- Prof. Neelkanth Kundu clarifies that the quantum state σ can be a two-by-two matrix for qubits, but it can be of arbitrary dimension for general quantum systems.

- The U matrix is a rectangular matrix due to the difference in dimensions between Alice's and the combined Bob-Eve systems.
- The concept of isometry and its difference from unitarity is explained, emphasizing that $U^\dagger U$ is identity for isometry, but $U U^\dagger$ is identity for unitarity.

Visualizing Isometric Matrices

- Paramshiv S and Brajesh Kumar seek examples to better understand isometric matrices.
- Prof. Neelkanth Kundu provides an example of a two-qubit isometric channel, V , which maps one qubit to two qubits.
- The dimensions of V are four-by-four for the output and two-by-two for the input.
- The isometry condition is verified by showing that $V^\dagger V$ is identity.

Calculating Private Capacity

- Prof. Neelkanth Kundu explains the formula for private capacity, which involves maximizing the mutual information between Alice and Bob minus the mutual information between Alice and Eve.
- The classical quantum state is represented by ρ_{XA} , which is a tensor product of classical and quantum states.
- The private capacity is the maximum over all quantum states and classical measurements.
- The example of the identity channel is used to illustrate the calculation of private capacity, showing that it is equal to the Holevo capacity for this channel.

Quantum Teleportation Protocol

- Prof. Neelkanth Kundu introduces the quantum teleportation protocol, which uses entanglement to transmit a quantum state from Alice to Bob.
- Alice and Bob share an entangled qubit, and Alice performs a Bell state measurement on her qubits.
- The outcomes of the Bell state measurement are transmitted to Bob through classical channels.
- Bob applies conditional operations based on the classical bits received from Alice to reconstruct the original quantum state.

Mathematical Derivation of Teleportation

- The mathematical derivation of the teleportation protocol is explained, showing how the quantum state at Alice's end is transformed and reconstructed at Bob's end.
- The importance of the Bell state measurement in collapsing the quantum state and the role of classical communication in transmitting the necessary information to Bob are emphasized.
- The protocol ensures that no cloning theorem is violated, and no faster-than-light communication occurs.
- The derivation includes steps to show how the quantum state at Bob's end is reconstructed based on the classical bits received from Alice.

Insights and Misconceptions

- Prof. Neelkanth Kundu addresses common misconceptions about quantum teleportation, such as the violation of no cloning theorem and faster-than-light communication.
- The importance of classical communication in the protocol is reiterated.
- The protocol ensures that the principles of quantum mechanics are preserved, and no information is leaked to the eavesdropper.
- The session concludes with a summary of the key points and an invitation for further study and exploration of the concepts discussed.